

Arju

Ph.D. Scholar-Climate Modeling with Machine Learning

Narela, Delhi - 110040

+91 9027183166

✉ arjujawla@nitdelhi.ac.in

in arju-990589192

Professional Summary

I am a researcher developing advanced multi-tasking machine learning frameworks for extended forecasts of wind-wave parameters along the Indian coastline. My work integrates deep learning, statistical modeling, and large-scale climate datasets (reanalysis, satellite, and buoy observations) to forecast significant wave height, wave period, wind speed, and coastal flooding risks. With a strong foundation in applied mathematics and data science, I focus on building scalable and interpretable AI-driven tools that can improve early-warning systems and support climate resilience for vulnerable coastal communities.

Current Position

Dec 2024–Present **Ph.D. Research Scholar** in Climate Modeling with Machine Learning, **National Institute of Technology, Delhi**

Aug 2024–Present **Junior Researcher**, Sponsored Project under OMAS (INCOIS)

Title: *"Extended Forecast of Wind-wave Parameters along the Indian Coastline using Multi-tasking Machine-learning Techniques"*

Objectives: Forecast wave parameters (SWH, swell, wind speed, etc.), estimate EAAP & EAFL, downscaling using ML, support coastal vulnerability mapping.

Research Interests

Climate Modeling, Machine Learning, Oceanography, AI for Earth Sciences, Applied Mathematics, Forecasting and Downscaling, Data Science.

Education

2024–Present **Ph.D. in Applied Science(Maths)**, NIT, Delhi, 9.3 CGPA

2021–2023 **M.Sc. Mathematics**, Hindu College, Delhi University, 66.43%

2018–2021 **B.Sc. Mathematics (Hons)**, Kurukshetra University, 86.42%

2018 **Intermediate**, CBSE, St. R C Convent School, 91%

2016 **High School**, CBSE, Moon Light Public School, 10 CGPA

Certifications

2024 **GATE MA (Mathematics)**, Score: 488, Rank: 393, Marks: 32

- 2024 **GATE DA (Data Science & Artificial Intelligence)**, *Score: 431, Rank: 3736 , Marks: 43*
- 2025 **Machine Learning & Deep Learning in python & R**, *Udemy*
- 2025 **FDP: Generative AI: Tehniques, tools and Applications**, *AICTE*
- 2023 **Google Advanced Data Analytics Certification**, *Coursera*
- 2023 **Python Complete Course for Beginners**, *Udemy*

Experience

- 2022–2023 **Subject Matter Expert (Mathematics)**, *Chegg*
- Provided detailed solutions to higher mathematics questions.
 - Assisted students with complex mathematical concepts and problems.

Skills

Programming	Python, C, C++, Matlab
Tools	GrADS, CDO, LaTeX, MS Suits
Data Formats	NetCDF, GRIB, HDF5 – Large Climate Dataset Handling
ML Frameworks	Tensorflow, Pytorch, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
Platforms	Linux/Unix, Jupyter, Kaggle
Mathematics	Statistics, Numerical Methods, Linear Algebra, Applied Math

Achievements

- 2024 Qualified GATE MA (Mathematics) and GATE DA (Data Science & Artificial Intelligence)
- 2021 Secured AIR 10 in DUET PG Entrance and AIR 6 in BHU PET
- Participated in and won various intercollege and state-level quizzes and competitions
- Awarded Best Student in school for academics, sports, and cultural activities
- 2015,2016,2017 Managed different House responsibilities as House Captain, Vice Captain, and Discipline Committees Member
- 2011, 2013 Got 1st prize in Kho-Kho at district level annual sports meet 2013 and 2nd prize in 2011

References

Dr. Prashant Kumar
 Department of Applied Sciences, NIT Delhi, India
 prashantkumar@nitdelhi.ac.in